

# Tiny ECG Classification on ESP32

A 23 KiB full-int8 CNN with 95.1% test accuracy and 52.09 ms ESP32 inference

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FROM-SCRATCH CNN / TFLITE MICRO

## Project Goal



Develop a **1D CNN from scratch** in NumPy to accurately classify different heartbeat types, apply **INT8 quantization** to reduce the model size, and optimize it for efficient deployment on an **ESP32**.

95.12%

INT8 TEST ACCURACY

73.9%

SMALLER MODEL

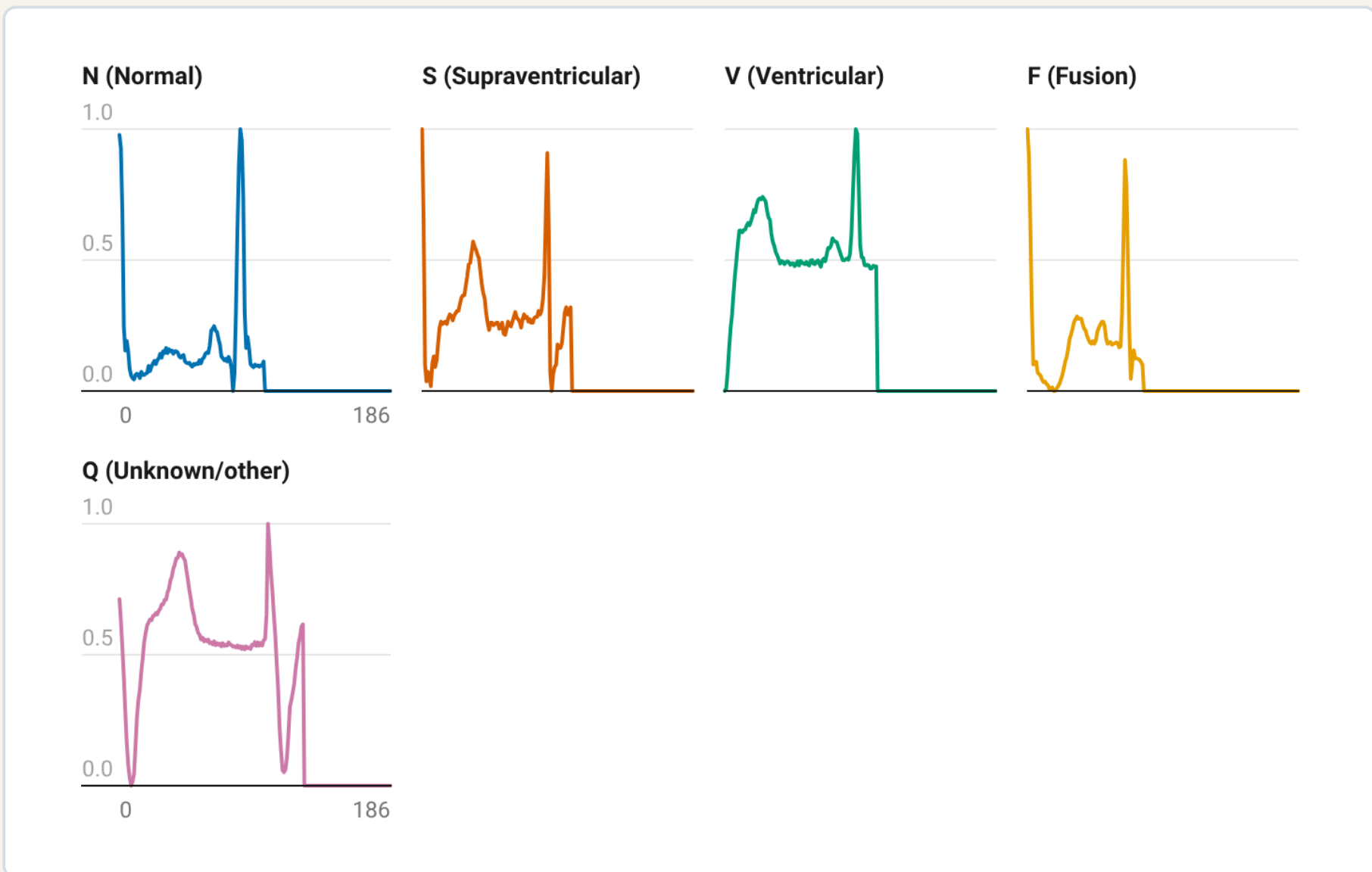
52.09 ms

ESP32 MEAN INFERENCE / BEAT

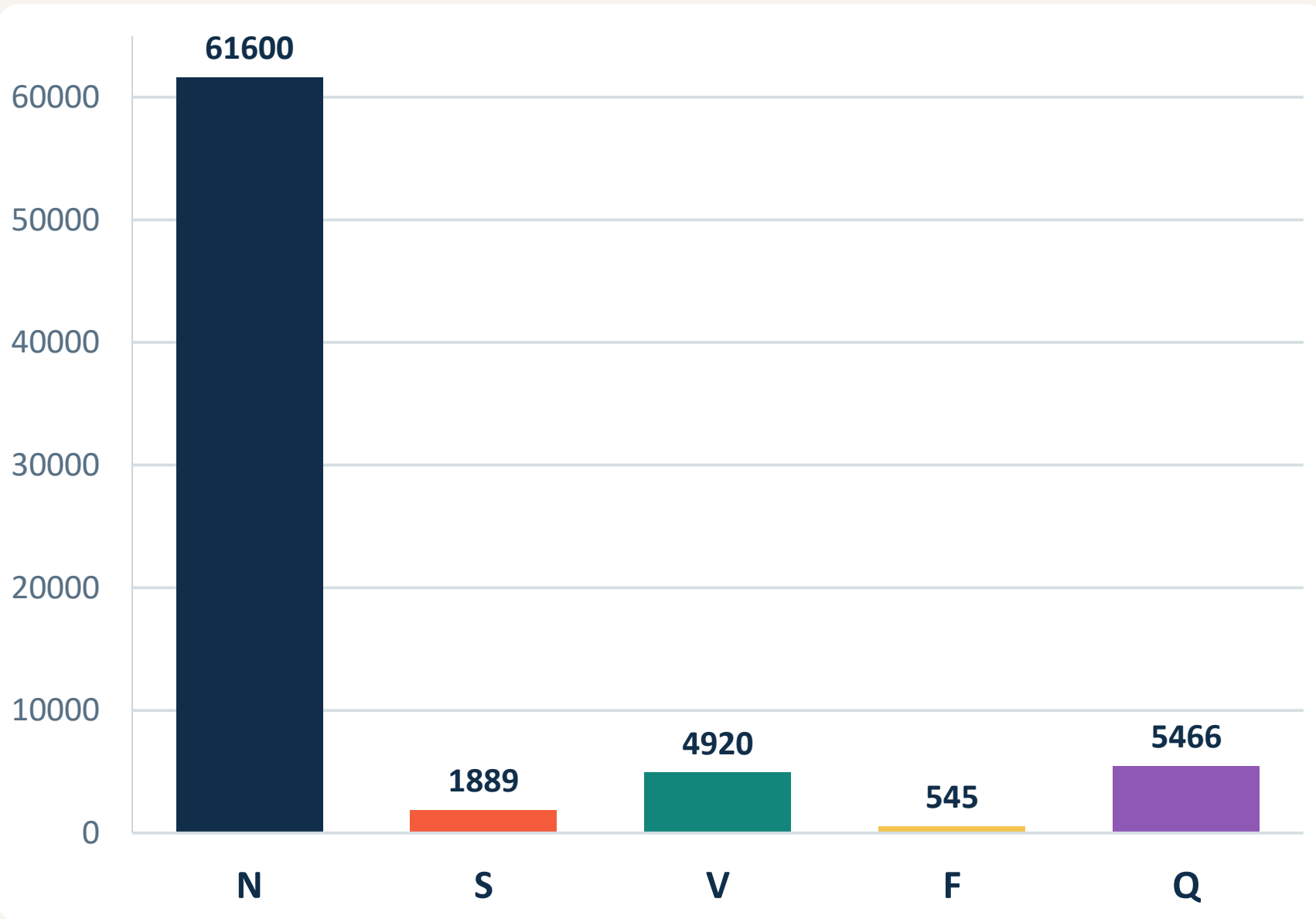
## 1. Heartbeat data are highly imbalanced

Five heartbeat types must be separated from 187-point ECG segments. Normal beats dominate training, so overall accuracy alone can hide minority-class failure.

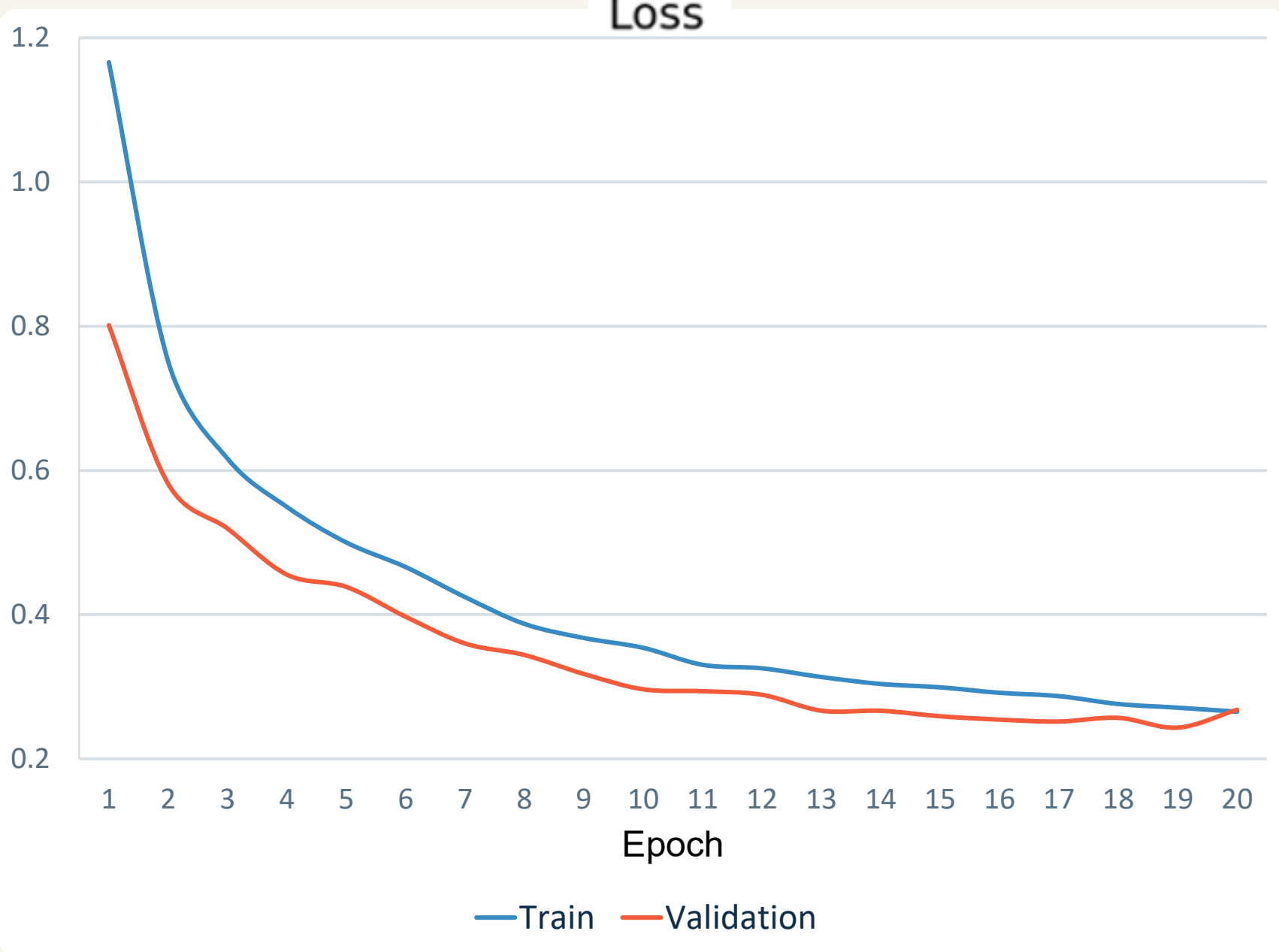
### Representative normalized beats



### Training class distribution



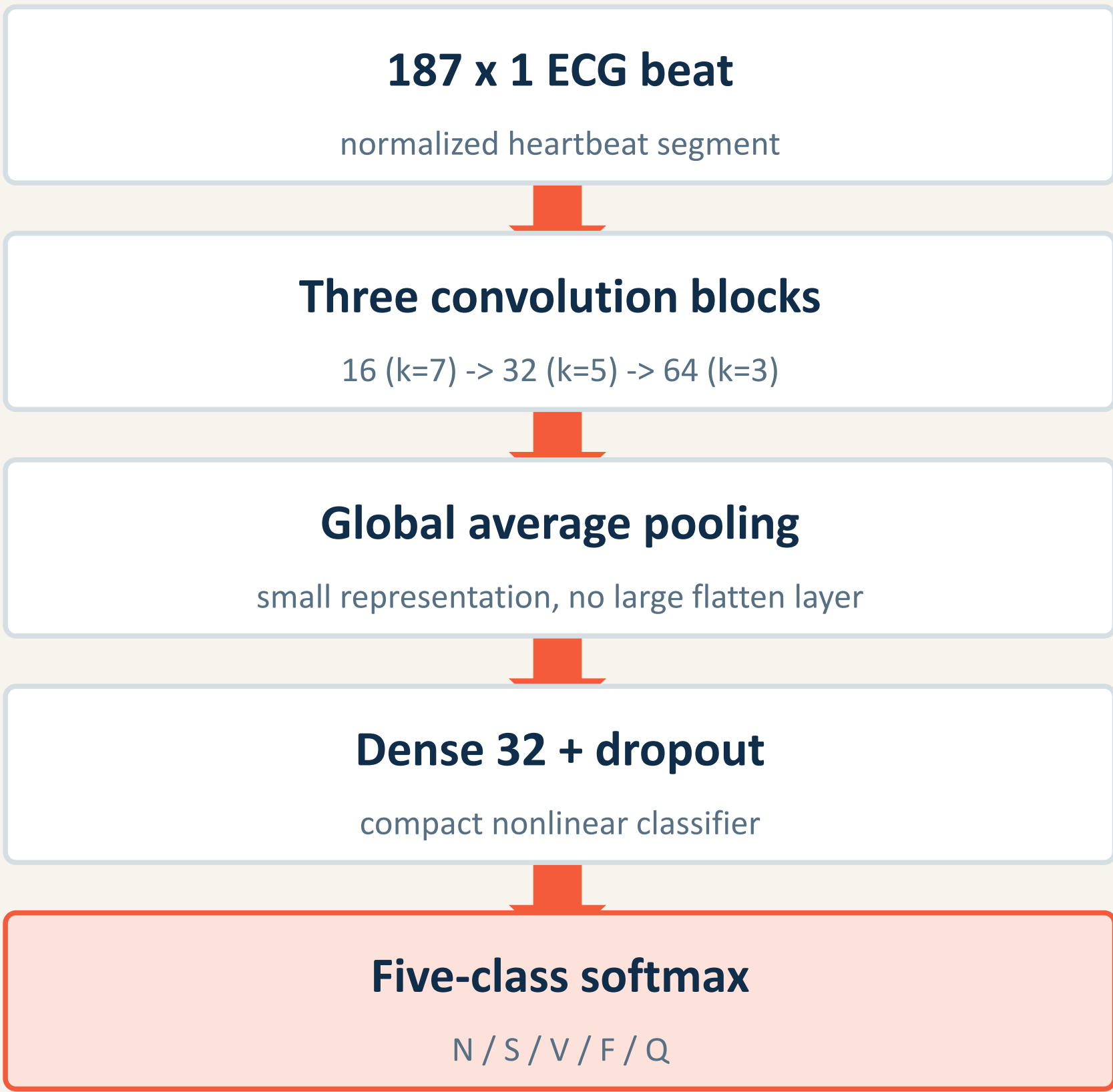
### Stable convergence without clear overfitting



## 2. Compact CNN, explicit imbalance controls

The complete forward pass, backward pass, weighted loss, and Adam updates are implemented in NumPy. Keras is used only as an export bridge.

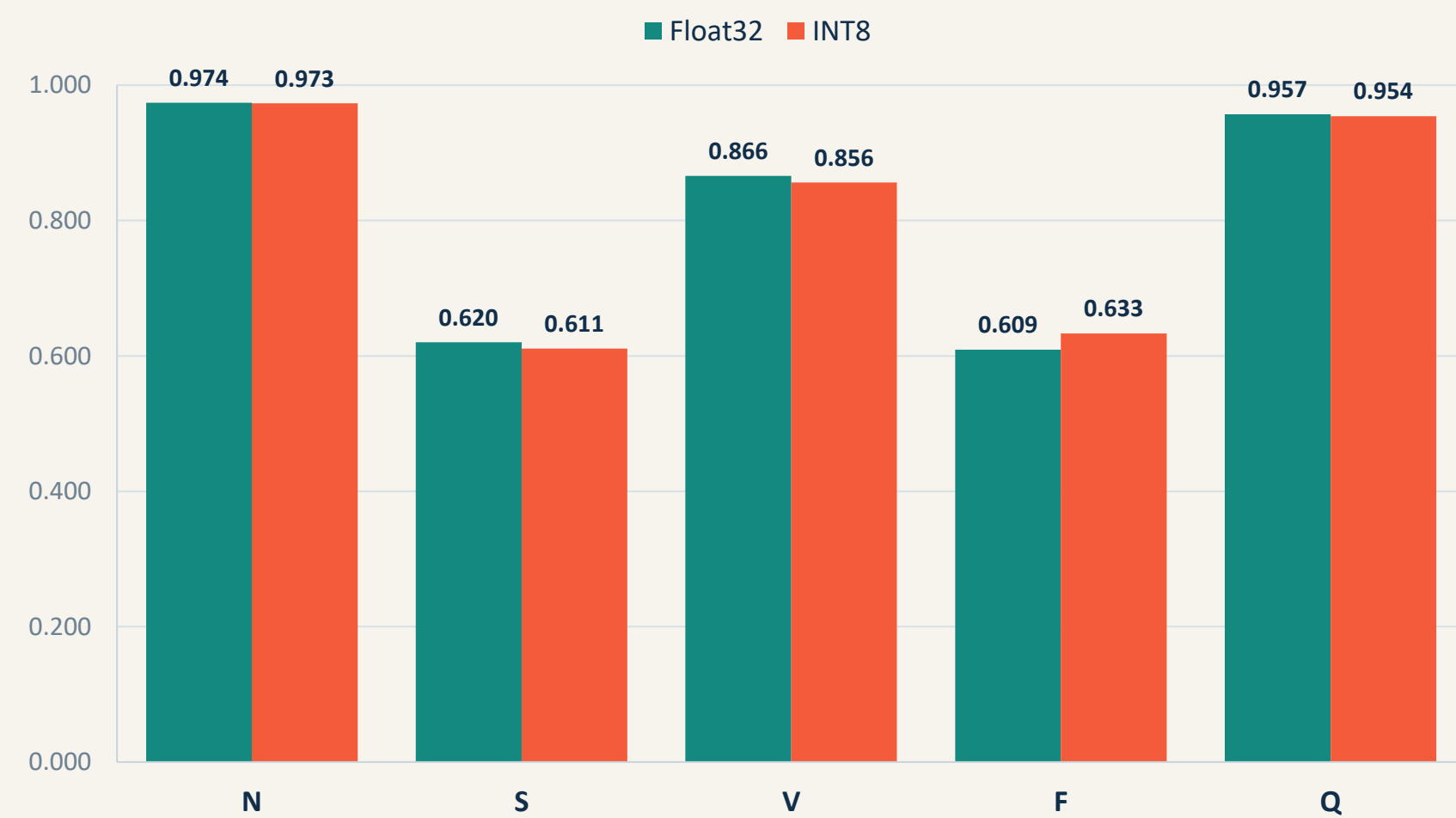
11,173 trainable parameters



### Three complementary strategies

- Capped class-weighted loss**  
 $w_c = \min(4, \frac{N}{5n_c})$  limits minority gradients.  
Rare errors matter without unstable minority gradients.
- Morphology-preserving augmentation**  
 $x'(t) = \alpha x(t + \Delta t) + b + \varepsilon(t)$   
 $\alpha$  := amplitude scaling,  $\Delta t$  := small time shift,  $b$  := baseline shift for rare beats,  $\varepsilon(t)$  := small noise  
Noise and small shifts preserve the ECG beat shape.
- Tested sampler: keep shuffle**  
Weighted and balanced mini-batches overcorrected.  
Natural shuffle retained 95.3% test accuracy.

### Quantization preserves per-class F1

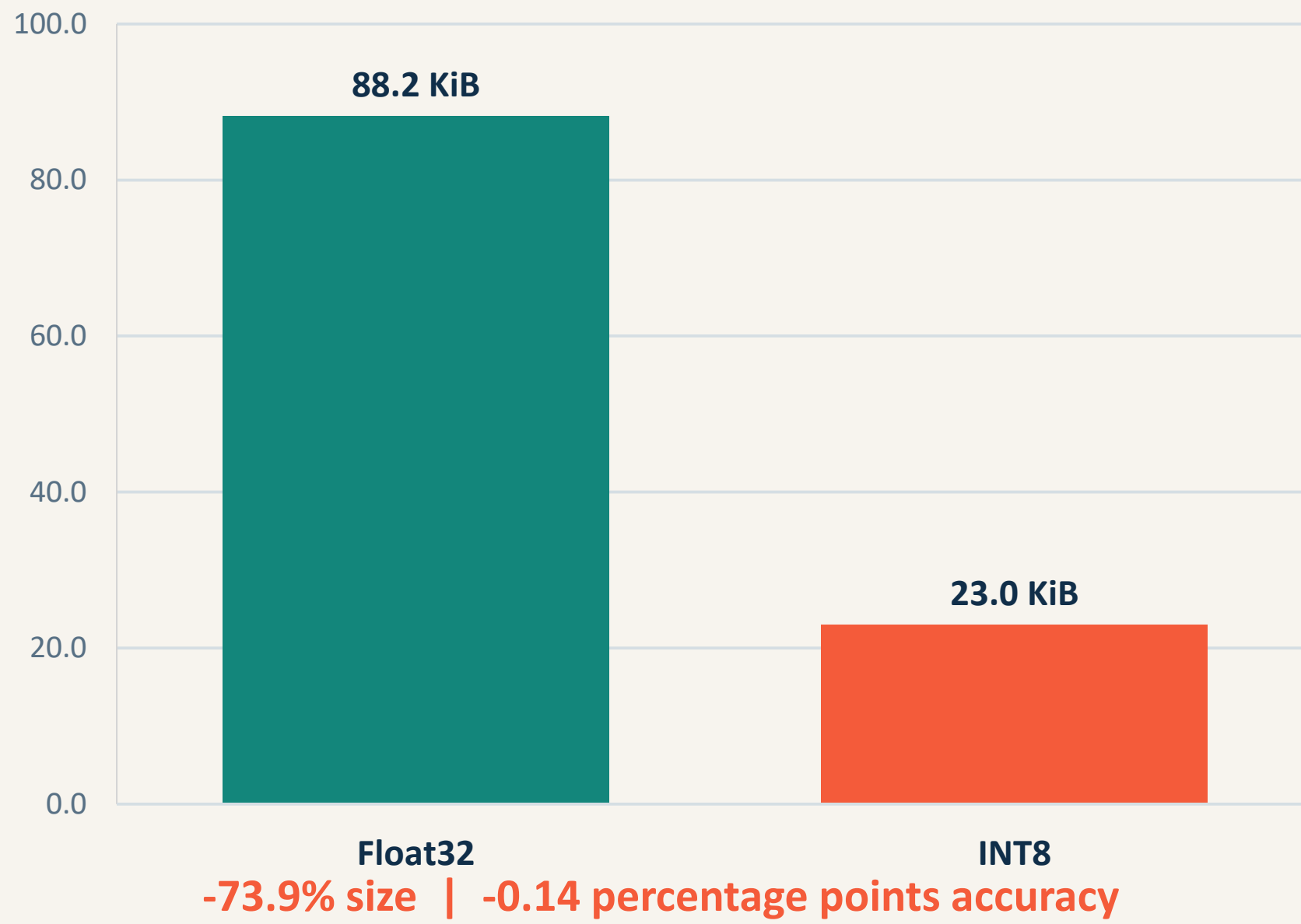


## 3. Full-int8 compression keeps model quality

Representative-data calibration quantizes weights, activations, inputs, and outputs to int8 for TensorFlow Lite Micro.

$$q = \text{clip}(\text{round}(\frac{x}{s} + z)); \quad x \approx s(q - z)$$

### Model Size



### INT8 normalized confusion matrix

|   | N    | S    | V    | F    | Q    |
|---|------|------|------|------|------|
| N | 0.96 | 0.01 | 0.02 | 0.01 | 0.00 |
| S | 0.37 | 0.56 | 0.04 | 0.01 | 0.01 |
| V | 0.02 | 0.00 | 0.95 | 0.02 | 0.00 |
| F | 0.10 | 0.00 | 0.09 | 0.81 | 0.00 |
| Q | 0.02 | 0.00 | 0.01 | 0.00 | 0.97 |

Predicted class

Class S remains hardest: 37% are confused with normal beats.

### Deployment path



### Prototype scope

Validated on stored test beats. Live ADC acquisition, heartbeat segmentation, patient-wise clinical validation, and confidence handling remain future work.

### TAKEAWAY

Full-int8 quantization is the deployment sweet spot: 23 KiB, 95.1% accuracy, and 52.09 ms ESP32 inference.